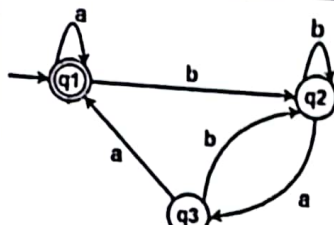
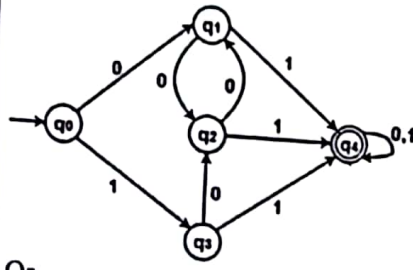


International Islamic University Chittagong Department of Computer Science and Engineering <i>B. Sc. in CSE Midterm Examination, Autumn 2022</i> Course Code: CSE 3527 Course Title: Compiler Total marks: 30 Time: 1 Hour and 30 minutes						
					CO	DL
1.				5	C01	C3
a)	Translate the following assignment statement step by step using compiler phases: $z = c/d + a*b$, here z, c, d are integer and a, b are floats.			5	C01	C2, C3
b)	Write the differences between lexical analyzer and parser. Identify the lexemes and their corresponding tokens and non-tokens from the following C code: <pre>#include<stdio.h> #define a 10 int sum(float x, float y) { float m; // this will add 2 numbers sum= x+y; if(x>y) printf("%d", sum); else return sum; }</pre>					
2.				4	C03	C1, C2
a)	What is context sensitive grammar? Identify the useless non terminals, Left Linear Grammar, Right Linear Grammar from the following grammar? $S \rightarrow ABd/a$ $A \rightarrow eBC/b$ $B \rightarrow aB/C$ $C \rightarrow aC/B$ Or, When do we call a grammar ambiguous? Check whether the given grammar G is ambiguous or not for the string "a(a)aa". $A \rightarrow AA$ $A \rightarrow (A)$ $A \rightarrow a$					
b)	Construct CFG for the language of all non-Palindromes. Or, Construct a CFG for the language $L = 0^n1^{4n}$ where $n \geq 1$.			2	C03	C3
c)	Write the regular expressions for the following languages: i. All strings of a's and b's with an even number of a's and an odd number of b's. ii. All strings of a's and b's that do not contain the substring abb.			4	C02	C3
3.						
a)	Find the regular expression for the following FA:			4	C02	C3

		
a)	Minimize the following DFA:  Or, Convert the regular expression $R = abb(a b)^*(a b)$ directly to its equivalent NFA	4 C02
b)	Describe the languages denoted by the following regular expressions: $a(a b)^*a$ $a^*ba^*ba^*ba^*$	6 / 13

International Islamic University Chittagong

Department of Computer Science and Engineering

B. Sc. in CSE, Midterm Exam, 5th Semester, Spring 2022

Course Code: CSE 3527

Total marks: 30

Course Title: Compiler

Time: 1 hours 30 minutes

[Answer all the questions; in some questions, there are options;
Figures in the right-hand margin indicate full marks.]

1. a. Specify each phase of a compiler explicitly with corresponding input and output while processing the following statement: 5

Expression := Term * factor + 128

- b. Consider the grammar: 5

$stmt \rightarrow if-stmt \mid other$
 $if-stmt \rightarrow if (exp) stmt$
 $\quad \quad \mid if (exp) stmt \text{ else } stmt$
 $exp \rightarrow 0 \mid 1$

- i. Now, show that the grammar is ambiguous by constructing two parse trees for the following string:
if (0) if (1) other else other

- ii. Rewrite the grammar so that it becomes unambiguous. 3

2. a. Consider the following grammar:

$E \rightarrow E \text{ or } T \mid T$
 $T \rightarrow T \text{ and } F \mid F$
 $F \rightarrow \text{not } F \mid (E) \mid \text{true} \mid \text{false}$

Now, construct a parse tree for the following sequence:

not (true or false)

- b. Suppose you are considering DFA from regular expression. Now, compute the 7
follow positions for the following regular expression:
(alb)*ab*ab

OR

Convert the regular expression $R = (1|0)^*1$ to its equivalent NFA using Thompson construction.

3. a. i. What are the advantages of separating the analysis phase of compiling into lexical analysis and parsing? 5
ii. Why context free grammar is preferred over regular expressions in parsing?
b. Convert the regular expression $R = (alb)^*abb(alb)$ directly to its equivalent DFA. 5

OR

Prove that $0^*(10)^*$ is equivalent to $(1^*0)^*1^*$



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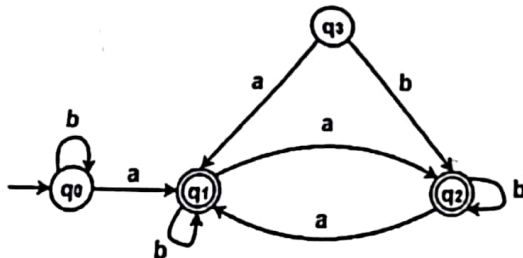
Department of Computer Science and Engineering

B. Sc. in CSE Midterm Examination, Autumn 2023

Course Code: CSE-3527 Course Title: Compiler

Total marks: 30 Time: 90 minutes

1. a. What is bytecode? How do Java language processors work? 2 CO1 C2
 b. How is an executable target program generated? Explain it using a language processing system. 3 CO1 C2
 c. Translate the following assignment statement step by step using compiler phases: 4 CO1 C3
 $z = a * b + c / 4;$ [$z, a, b,$ and c contain floating point values]
2. a. Consider the CFG: $S \rightarrow a + SS \mid * SS \mid S \mid a$ 2 CO1 C1
 1. Show that: aa^+a^* can be generated by the grammar. $\vdash a^+aa$ C2
 2. Draw the parse tree
 b. Consider the grammar: $P \rightarrow aPbP \mid bPaP \mid \epsilon$ 4 CO1 C3
 Consider $ababab$ as the string to check the ambiguity of the given grammar.
 c. Construct the regular expressions and provide two accepted strings for each. 4 CO2 C3
 (Any two):
 1) $\{w \mid w: \text{starts with a 'b' and has at least one a}\}$
 2) $\{w \mid w: \text{does not contain the substring abab}\}$
 3) $\{w \mid w: \text{set of strings consisting of even number of a's followed by b}\}$
 4) $\{w \mid w: \text{set of strings of 1's and 0's in even length}\}$
3. a. Minimize the following DFA: 4 CO2 C3



- b. Convert the following regular expression directly to its equivalent NFA: 2 CO2 C3
 $abb(a|b)^*ba$
 OR,
 $(a|b)^*a(a|b)(a|b)$
 c. Convert the following NFA to DFA: 4 CO1 C3

